Priyanshu Macwan

Computer Science & Engineering Student · Full-Stack Developer · Data Science Enthusiast

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PROFESSIONAL SUMMARY

Second-year Computer Science & Engineering student with hands-on experience building full-stack web applications, IoT systems, and data-driven tools. Independently shipped 5+ complete projects spanning React, FastAPI, Firebase, and ESP32 embedded hardware. Currently exploring Data Science and Machine Learning fundamentals with a focus on practical application. Preparing for GATE-CS with a consistent, self-driven approach to learning and problem solving.

EDUCATION

Bachelor of Technology — Computer Science & Engineering 2024 – 2028 (Expected)

Gujarat Technological University, Gujarat, India 2nd Year · Semester 4

Relevant Coursework: Data Structures & Algorithms, Digital Electronics, Object-Oriented Programming, Database Management Systems, Discrete Mathematics, Computer Networks, Operating Systems

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, Python, HTML5, CSS3
Frameworks & Tools: React.js, FastAPI, Node.js, Tailwind CSS, REST APIs
Databases & Cloud: Firebase Realtime DB, Firestore, Google Sheets API
Embedded Systems: ESP32 DOIT DevKit V1, Fingerprint Sensors, Arduino IDE
Dev Tools: Git, GitHub, VS Code, Tinkercad, npm, Postman
Data Science: NumPy, Pandas, Matplotlib, Scikit-learn (learning), Jupyter Notebooks
Concepts: OOP, DSA, REST Architecture, PWA, Agile, UI/UX Principles

PROJECTS

Space Station Storage Management System 2024

React PWA · FastAPI · Firebase Realtime Database · Python Full-Stack | Personal Project

- Built a Progressive Web App simulating inventory management for a space station, with real-time data sync via Firebase Realtime Database.
- Designed a FastAPI backend with RESTful CRUD endpoints; integrated dynamic dashboards to visualize storage capacity, usage trends, and item status.
- Added multi-field search, filtering, and sort; automated PDF and CSV report generation for operational inventory summaries.

Biometric Online Attendance System 2024

ESP32 DOIT DevKit V1 · Fingerprint Sensor · Firebase · Google Sheets API IoT + Cloud | Personal Project

- Developed an IoT attendance solution using an ESP32 microcontroller and I Jager fingerprint scanner for contactless biometric authentication.
- Implemented a subject-selection workflow before check-in; synced records to Firebase Realtime Database for multi-device cloud access.
- Automated real-time export to Google Sheets via the Sheets API, enabling teacher dashboards with live attendance tracking and downloadable reports.

Blockchain-Based Athlete Management System 2024

Blockchain Architecture · Decentralized Design · Hackathon Hackathon Project

- Designed a decentralized platform for tracking athlete lifecycle data — performance records, contracts, and career milestones — using blockchain-inspired immutability.
- Identified an underexplored hackathon domain to maximize technical differentiation; structured the problem statement for originality and competitive advantage.

PROJECTS (CONTINUED)

ReWear Platform — Frontend Development
2024

React.js · HTML5 · CSS3 · Component Architecture
Web Development | Personal Project

- Built a multi-page React frontend for a sustainable fashion resale platform, covering login/register flow, a user dashboard, a browse section, and an add-item form.
- Implemented shared CSS architecture and component-based design for consistent styling and scalability; applied mobile-first layout principles throughout.
- Focused on clean UI/UX patterns — card layouts, interactive state transitions — to deliver a polished, recruiter-grade interface.

Digital Electronics & ALU Design
2024

Tinkercad · Logic Gates · Digital Circuit Simulation
Academic + Practical

- Designed and simulated combinational and sequential logic circuits in Tinkercad, including half-adder, full-adder, MUX, and flip-flops.
- Built a functional ALU prototype supporting ADD, SUB, AND, OR, and XOR operations using standard gate-level design — directly reinforcing GATE-CS coursework.

Animated Birthday Wish Website
2024

HTML5 · CSS3 Animations · Vanilla JavaScript
Creative Development

- Created a fully animated celebration webpage with sequential reveal animations, CSS fireworks, floating balloons, and interactive messaging — a personal project for Badi Behena.
- Demonstrated clean use of CSS keyframes, transition timing, and JS DOM sequencing for a memorable, lightweight user experience.

ACHIEVEMENTS & HACKATHONS

- Hackathon Competitor — Submitted original projects in college-level and inter-college hackathons; consistently chose unique problem domains to improve competitive positioning.
- Independent Builder — Shipped 5+ complete real-world projects spanning IoT, full-stack web, blockchain architecture, and creative frontend without guided coursework.
- GATE-CS Aspirant — Proactively preparing for the Graduate Aptitude Test in Engineering (GATE-CS) alongside 2nd-year coursework, covering DSA, Discrete Maths, and Digital Electronics.
- Self-Driven Learner — Actively studying Data Science and ML fundamentals (Python data stack, NumPy, Pandas, Scikit-learn) outside formal curriculum.

CERTIFICATIONS & CONTINUOUS LEARNING

- Actively studying Python for Data Analysis — NumPy, Pandas, Matplotlib — through structured self-directed curriculum.
- Pursuing Machine Learning fundamentals: supervised learning, regression, classification, model evaluation, and feature engineering.
- Hands-on experience integrating real-world APIs (Google Sheets API, Firebase REST, FastAPI endpoints) across multiple shipped projects.
- Studying clean code principles, OOP design patterns, and software engineering fundamentals through self-study and iterative project building.
- Practicing DSA problems regularly on platforms like LeetCode and GeeksforGeeks to strengthen algorithmic thinking for GATE and technical interviews.

CURRENT FOCUS — 2025

Data Science & Machine Learning *— Building proficiency in the Python data stack and core ML algorithms; targeting Data Science internships.*
Analytical Problem Solving *— Solving DSA problems daily to sharpen algorithmic thinking for GATE and software engineering interviews.*
Full-Stack Development *— Deepening React and FastAPI skills; exploring TypeScript and Docker for more production-ready builds.*
Portfolio & Open Source *— Planning first open-source contributions; expanding GitHub portfolio with diverse, well-documented projects.*

INTERESTS

Space Science & Aerospace · **Embedded Systems & IoT** · **Data Science & AI** · **Competitive Hackathons** · **Open Source** · **Digital Electronics** · **Scientific Computing** · **Web Development**